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Sound extraction system and sound extraction method

Abstract

To provide a sound extraction system and a sound extraction method capable of accurately extracting, from mixture signals, a signal corresponding to a sound which a user wants to extract. The sound extraction system includes a sound extraction device configured to extract, from mixture signals including a signal corresponding to an extraction target sound, the signal corresponding to the extraction target sound. The sound extraction device is configured to extract the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and a text representing a range of the extraction target sound.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a sound extraction system and a sound extraction method.

2. Description of the Related Art

(2) It is important to extract a sound having a specific feature from sounds in which a plurality of sound sources are mixed. For example, sounds are recorded, and an abnormality of a facility or a machine or a sign thereof is automatically detected from a sound (abnormal sound). However, environmental noise may be loud, and accuracy of abnormal sound detection in such a case may be significantly reduced. Therefore, in order to improve the accuracy of the abnormal sound detection and analyze the abnormal sound itself, it is required to extract (also referred to as emphasize) a target sound of the facility or the machine from the recorded sounds.

(3) A surrounding situation also is recognized based on sounds recorded by a microphone of a monitoring camera, a drive recorder, a watching robot, a smart speaker, and the like. However, environmental noise may still be loud, and accuracy of situation recognition in such a case may be significantly reduced. Therefore, in order to improve the accuracy of the situation recognition and analyze the recorded sounds, it is required to extract a sound serving as a clue for the situation recognition from the recorded sounds.

(4) Regarding extraction of a sound, for example, techniques disclosed in JP-A-2014-178886 (Patent Literature 1) and Y. Sudo, “Environmental sound segmentation utilizing Mask U-Net,” in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019 (Non-Patent Literature 1) are known. Non-Patent Literature 1 discloses that “The waveforms of mixed environmental sounds are transformed into spectrogram. By using this input, Mask U-Net consists of sound event detection CNN and segmentation U-Net predicts masks for separating out each class from the input spectrogram. Inverse SIFT is applied to reconstruct the time domain signal.”. In the technique disclosed in Non-Patent Literature 1 (hereinafter, also referred to as “usual technique”), sounds are classified into a finite number of types of events defined in advance, and a sound is extracted for each type of event.

(5) However, in the related art, when a range of a sound which a user wants to extract does not match a type of event defined in advance, a sound which the user wants to extract cannot be extracted. As an example, in the related art, it is not possible to extract a sound with a granularity finer than that of a type of event defined in advance. For example, even if a type of event is defined as a sound of hitting a metal, the sound of hitting a metal may include a sound of “bang” and a sound of “clang-clang”. Therefore, for example, when the user wants to extract the sound of “bang” as the “sound of hitting a metal”, in the related art, the sound of “clang-clang” may be extracted as the sound of hitting a metal by classification of an event of the “sound of hitting a metal”.

(6) Thus, in the related art, a sound which the user wants to extract may not be accurately extracted. The invention is made to solve the above problems. That is, one object of the invention is to provide a sound extraction system and a sound extraction method capable of accurately extracting a signal corresponding to a sound which a user wants to extract from mixture signals.

SUMMARY OF THE INVENTION

(7) In order to solve the above problems, a sound extraction system according to the invention includes a sound extraction device configured to extract, from mixture signals including a signal corresponding to an extraction target sound, the signal corresponding to the extraction target sound, and the sound extraction device is configured to extract the signal corresponding to the extraction

target sound from the mixture signals based on the mixture signals and a text representing a range of the extraction target sound.

(8) A sound extraction method according to the invention is performed using a sound extraction device configured to extract, from mixture signals including a signal corresponding to an extraction target sound, the signal corresponding to the extraction target sound, and the sound extraction method includes: extracting, by the sound extraction device, the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and a text representing a range of the extraction target sound.

(9) According to the invention, a signal corresponding to a sound which a user wants to extract can be accurately extracted from mixture signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a first embodiment of the invention.

(2) FIG. 2 is a block diagram showing a configuration example of an information processing device.

(3) FIG. 3 is a block diagram showing a configuration example of a learning subsystem for each function.

(4) FIG. 4 is a flowchart showing an example of a processing flow of the learning subsystem.

(5) FIG. 5 is a block diagram showing a configuration example of a sound extraction subsystem for each function.

(6) FIG. 6 is a flowchart showing an example of a processing flow of the sound extraction subsystem.

(7) FIG. 7 is data showing an example of extraction results of the sound extraction system.

(8) FIG. 8 is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a second embodiment of the invention.

(9) FIG. 9 is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a third embodiment of the invention.

(10) FIG. 10 is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a fourth embodiment of the invention.

(11) FIG. 11 is a block diagram showing a configuration example of the learning subsystem for each function.

(12) FIG. 12 is a flowchart showing an example of a processing flow of the learning subsystem.

(13) FIG. 13 is a block diagram showing a configuration example of the sound extraction subsystem for each function.

(14) FIG. 14 is a flowchart showing an example of a processing flow of the sound extraction subsystem.

(15) FIG. 15 is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a fifth embodiment of the invention.

(16) FIG. 16 is a block diagram showing a configuration example of the learning subsystem for each function.

(17) FIG. 17 is a flowchart showing an example of a processing flow of the learning subsystem.

(18) FIG. 18 is a block diagram showing a configuration example of the sound extraction subsystem for each function.

(19) FIG. 19 is a flowchart showing an example of a processing flow of the sound extraction subsystem.

(20) FIG. 20 is a block configuration diagram showing a schematic configuration example of a

sound extraction system according to a sixth embodiment of the invention.

(21) FIG. **21** is a block configuration diagram showing a schematic configuration example of a sound extraction system according to a seventh embodiment of the invention.

(22) FIG. **22** is a block configuration diagram showing a schematic configuration example of a sound extraction system according to an eighth embodiment of the invention.

(23) FIG. **23** is a block diagram showing a configuration example of the learning subsystem for each function.

(24) FIG. **24** is a flowchart showing an example of a processing flow of the learning subsystem.

(25) FIG. **25** is a block diagram showing a configuration example of the sound extraction subsystem for each function.

(26) FIG. **26** is a flowchart showing an example of a processing flow of the sound extraction subsystem.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(27) Hereinafter, a sound extraction system according to embodiments of the invention will be described with reference to the drawings and the like.

First Embodiment

SUMMARY OF INVENTION

(28) FIG. **1** is a block configuration diagram showing a schematic configuration example of a sound extraction system **100** according to a first embodiment of the invention. As shown in FIG. **1**, the sound extraction system **100** includes a learning subsystem **110**, a feature extraction model database **120**, a text-embedded extraction model database **130**, a time-frequency mask generation model database **140**, a sound extraction subsystem **150**, and a training data set database **160**. The learning subsystem **110** may also be referred to as a “learning device”. The sound extraction subsystem **150** may also be referred to as a “sound extraction device”.

(29) First, the sound extraction system **100** reads out, from the training data set database **160**, a set (also referred to as a “training data set”) of three items including a target-signal (a signal corresponding to an extraction target sound) time waveform, a “mixture-signal time waveform obtained by mixing the target-signal time waveform (target signal) and noise other than the extraction target sound (a time waveform of a signal corresponding to noise (a signal corresponding to noise))”, and a variable-length onomatopoeia text (an onomatopoeia text corresponding to the extraction target sound), and inputs the set to the learning subsystem **110**.

(30) Here, instead of reading out the “mixture-signal time waveform obtained by mixing the target-signal time waveform (target signal) and noise other than the extraction target sound (a time waveform of a signal corresponding to noise (a signal corresponding to noise))” from the training data set database **160**, by reading out the “noise other than the extraction target sound (a time waveform of a signal corresponding to noise (a signal corresponding to noise))” before mixing and mixing the noise other than the extraction target sound with the target-signal time waveform (the signal corresponding to the extraction target sound), the “mixture-signal time waveform obtained by mixing the target-signal time waveform (target signal) and noise other than the extraction target sound (a time waveform of a signal corresponding to noise (a signal corresponding to noise))” may be generated, and the set of three items may be generated. Mixing the signals after the noise is read out has two advantages. One advantage is that by mixing signals at a signal-to-noise ratio assumed when the sound extraction system **100** is used to create a training data set, a model can be trained such that extraction suitable for a signal-to-noise ratio for each use scene can be performed. The other advantage is that a storage capacity required for the training data set database **160** can be reduced.

(31) A set of three items generated by mixing after reading from the training data set database **160** is also the set of three items and is hereinafter referred to as the set of three items or the training data set.

(32) The learning subsystem **110** executes learning processing based on the set of three items,

outputs a feature extraction model, a text-embedded extraction model, and a time-frequency mask generation model, and stores the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model in the respective databases. That is, the learning subsystem **110** stores the feature extraction model in the feature extraction model database **120**, stores the text-embedded extraction model in the text-embedded extraction model database **130**, and stores the time-frequency mask generation model in the time-frequency mask generation model database **140**.

(33) The sound extraction subsystem **150** reads out the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model from the databases (the feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140**), and executes sound extraction processing based on (using) the read feature extraction model, text-embedded extraction model, and time-frequency mask generation model. Accordingly, the sound extraction subsystem **150** extracts an extraction-signal time waveform from the mixture-signal time waveform and the variable-length onomatopoeia text. Further, the sound extraction subsystem **150** outputs the extraction-signal time waveform.

(34) With such a basic configuration, even when a range of a sound which a user wants to extract as a certain type of event cannot be strictly defined in advance, the sound extraction system **100** can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to the sound which the user wants to extract.

(35) A definition of a type of event as used in Non-Patent Literature 1 described above differs depending on an application site, and it is rare that the range of the sound which the user wants to extract matches a type of event defined in advance. When the range of the sound which the user wants to extract does not match the type of event defined in advance, a sound which the user wants to extract cannot be extracted. On the other hand, since onomatopoeia has relatively high versatility, there is a high possibility that the onomatopoeia can be used across the application site.

(36) Further, as processing of outputting an environmental sound in response to an input of onomatopoeia, a technique disclosed in Patent Literature 1 described above is known. Patent Literature 1 discloses that “a speech input unit configured to receive a speech signal; a speech recognition unit configured to perform speech recognition processing on the speech signal input to the speech input unit and to generate onomatopoeia; a sound data storage unit which stores an environmental sound and onomatopoeia corresponding to the environmental sound; a correlation information storage unit which stores correlation information in which first onomatopoeia, second onomatopoeia, and a frequency of selecting the second onomatopoeia when the first onomatopoeia is recognized by the speech recognition unit are correlated with each other; a conversion unit configured to convert the first onomatopoeia recognized by the speech recognition unit into the second onomatopoeia corresponding to the first onomatopoeia using the correlation information; and a retrieval and extraction unit configured to extract the environmental sound corresponding to the second onomatopoeia converted by the conversion unit from the sound data storage unit and to rank and present a plurality of candidates of an extracted environmental sound based on frequencies of selecting the plurality of candidates of the extracted environmental sound are included”.

(37) However, the technique of Patent Literature 1 cannot extract a sound from mixture sounds. “Extract” in Patent Literature 1 means to perform retrieval from a database and retrieve an environmental sound which matches a condition. The technique of Patent Literature 1 has only a mapping from an environmental sound to onomatopoeia, and does not have a mapping from onomatopoeia to an environmental sound, and therefore, an output sound is only a sound present in the database. Unless the mixture sounds exactly the same as the sound present in the database are input, the sound cannot be extracted from the mixture sounds. In extraction of the sound from the mixture sounds, which is a problem of the invention, it is almost impossible to input the mixture

sounds exactly the same as the sound present in the database. Therefore, in Patent Literature 1, the sound cannot be extracted from the mixture sounds.

(38) <Hardware Configuration>

(39) The sound extraction system **100** can be implemented by, for example, a computer (information processing device). FIG. 2 is a block diagram showing a configuration example of the information processing device. As shown in FIG. 2, an information processing device **200** includes a CPU **201**, a ROM **202**, a RAM **203**, a non-volatile storage device (HDD) **204** from and to which data read and write can be performed, a network interface **205**, an input/output interface **206**, and the like. These components are communicably connected to one another via a bus **207**. The CPU **201** loads various programs (not shown) stored in the ROM **202** and/or the storage device **204** into the RAM **203**, and executes the programs loaded into the RAM **203**, thereby implementing various functions. As described above, the various programs executed by the CPU **201** are loaded into the RAM **203**, and data used when the CPU **201** executes the various programs is temporarily stored in the RAM **203**. The ROM **202** and/or the storage device **204** are non-volatile storage media, and stores various programs. The network interface **205** is an interface for connecting the information processing device **200** to a network. The input/output interface **206** is an interface to be connected to an operation device such as a keyboard and a mouse, an acoustic device such as a microphone, and a display device such as a display.

(40) For example, the training data set database **160** of the sound extraction system **100** is implemented by a database stored in the storage device **204** in the information processing device **200**. The learning subsystem **110** of the sound extraction system **100** is implemented by the information processing device **200**. The feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140** of the sound extraction system **100** are implemented by databases stored in the storage device **204** in the information processing device **200**. The sound extraction subsystem **150** of the sound extraction system **100** is implemented by the information processing device **200**. The information processing device **200** constituting one system or the like may be a plurality of information processing devices or a virtual information processing device constructed on a cloud.

(41) <Learning Subsystem>

(42) (Function of Learning Subsystem)

(43) Hereinafter, a configuration of the learning subsystem **110** will be described mainly for each function. FIG. 3 is a block diagram showing a configuration example of the learning subsystem **110** for each function. As shown in FIG. 3, the learning subsystem **110** includes a target-signal frame division processing unit **111**, a target-signal window function multiplication unit **112**, a target-signal frequency domain signal generation unit **113**, a mixture-signal frame division processing unit **114**, a mixture-signal window function multiplication unit **115**, a mixture-signal frequency domain signal generation unit **116**, a feature extraction unit **117**, a phoneme conversion unit **118**, a text-embedded extraction unit **119**, a time-frequency mask generation unit **119a**, a time-frequency mask multiplication unit **119b**, and a learning unit **119c**. The target-signal frame division processing unit **111**, the target-signal window function multiplication unit **112**, the target-signal frequency domain signal generation unit **113**, the mixture-signal frame division processing unit **114**, the mixture-signal window function multiplication unit **115**, the mixture-signal frequency domain signal generation unit **116**, the feature extraction unit **117**, the phoneme conversion unit **118**, the text-embedded extraction unit **119**, the time-frequency mask generation unit **119a**, the time-frequency mask multiplication unit **119b**, and the learning unit **119c** are implemented by various programs (not shown) stored in the ROM **202** and/or in the storage device **204** of the information processing device **200**.

(44) The target-signal frame division processing unit **111** performs frame division of a target-signal time waveform **D10**, and outputs a target-signal frame division signal (not shown). The target-signal window function multiplication unit **112** performs window function multiplication, and

converts the target-signal frame division signal into a target-signal window function multiplication signal (not shown).

(45) The target-signal frequency domain signal generation unit **113** performs short-time Fourier transform, and converts the target-signal window function multiplication signal into a target-signal time-frequency domain representation **D11**. The target-signal frequency domain signal generation unit **113** may use a frequency transform method such as “constant Q transform (CQT)” instead of the short-time Fourier transform.

(46) The mixture-signal frame division processing unit **114** performs frame division of the mixture-signal time waveform **D20**, and outputs a mixture-signal frame division signal (not shown).

(47) The mixture-signal window function multiplication unit **115** performs window function multiplication, and converts the mixture-signal frame division signal into a mixture-signal window function multiplication signal (not shown). The mixture-signal frequency domain signal generation unit **116** performs short-time Fourier transform, and converts the mixture-signal window function multiplication signal into a mixture-signal time-frequency domain representation **D21**. The mixture-signal frequency domain signal generation unit may use the frequency transform method such as the “constant Q transform (CQT)” instead of the short-time Fourier transform.

(48) The feature extraction unit **117** converts the mixture-signal time-frequency domain representation **D21** into a sound feature vector **D22**. In the present example, the feature extraction unit **117** uses a feature extraction model which is a neural network having a variable weighting factor parameter. The feature extraction unit **117** inputs the mixture-signal time-frequency domain representation **D21** to a latest feature extraction model updated immediately before, and outputs the sound feature vector **D22**. The feature extraction model may be, for example, a neural network in which a plurality of convolution layers, activation functions, and pooling layers are stacked with a skip connection interposed thereamong.

(49) The sound feature vector **D22** may be an amplitude spectrogram of the time-frequency domain representation **D21**. In this case, the feature extraction unit **117** calculates an amplitude spectrogram (vector) of the time-frequency domain representation **D21**. In this case, the weighting factor parameter used for the calculation is invariable. For example, the sound feature vector may be a power spectrogram of the time-frequency domain representation **D21**. In this case, the feature extraction unit **117** calculates a power spectrogram (vector) of the time-frequency domain representation **D21**. In this case, the weighting factor parameter used for the calculation is invariable. For example, the sound feature vector **D22** may be a logarithmic mel power spectrogram of the time-frequency domain representation **D21**. In this case, the feature extraction unit **117** calculates a power spectrogram of the time-frequency domain representation **D21**, calculates a mel power spectrogram by multiplying the obtained power spectrogram by a mel filter bank, and performs logarithm on the obtained mel power spectrogram, thereby outputting a logarithmic mel power spectrogram (vector). In this case, the weighting factor parameter used for the calculation is invariable. A filter bank such as a $\frac{1}{3}$ octave band filter may be used instead of the mel filter bank.

(50) Further, the sound feature vector **D22** may be a time series of a mel frequency cepstrum coefficient (MFCC) instead of the logarithmic mel power spectrogram. In that case, the feature extraction unit **117** calculates a logarithmic value of the power spectrogram, multiplies the logarithmic value by the filter bank, performs discrete cosine transform, and outputs a time series (vector) of the MFCC. In this case, the weighting factor parameter used for the calculation is invariable.

(51) The sound feature vector **D22** may be a time difference with respect to a logarithmic mel power spectrogram or a time difference of the MFCC, a time series (delta) of time differentiation, or a concatenated vector thereof. In any of these cases, the weighting factor parameter used for the calculation is invariable.

(52) The phoneme conversion unit **118** outputs a variable-length phoneme string **D31** from a

variable-length onomatopoeia text **D30** by phoneme conversion processing. For example, when the onomatopoeia text **D30** is “clang-clang”, “/kaNkaN/” is output as the phoneme string **D31**. When the onomatopoeia text **D30** is “clang-clang boom”, “/katakata do:N/” is output as the phoneme string **D31**.

(53) The text-embedded extraction unit **119** outputs, using a latest text-embedded extraction model, a text-embedded vector **D32** from the phoneme string **D31**. The text-embedded vector **D32** is a vector having a predetermined number of dimensions D_s . First, the text-embedded extraction unit **119** assigns a one-hot vector to each phoneme in the input phoneme string **D31** to obtain a one-hot vector sequence. The one-hot vector here is a vector in which 1 is assigned only to a dimension corresponding to a type of phoneme (“/a/, /i/, /u/, /e/, /o/, /k/, /s/, /N/”, and the like) as a conversion target, and 0 is assigned to the other dimensions. Next, the one-hot vector sequence is input to the text-embedded extraction model, and the text-embedded vector **D32** is output. The text-embedded extraction model may be a known Transformer model, or a recurrent neural network with layers such as long-short-term-memory (LSTM), bidirectional LSTM, gated recurrent unit (GRU), and bidirectional GRU.

(54) The time-frequency mask generation unit **119a** generates, using a latest time-frequency mask generation model, a time-frequency mask from the sound feature vector **D22** and the text-embedded vector **D32**.

(55) The time-frequency mask is an estimate of what ratio of amplitudes of the mixture signals is an extraction signal at each time-frequency in a time-frequency domain representation. That is, the time-frequency mask takes a value larger than 0 and smaller than 1 at each time-frequency. The closer to 1, the larger portion of the amplitudes of the mixture signals are the extraction signal, and the closer to 0, the larger portion of the amplitudes of the mixture signals are components other than the extraction signal.

(56) The time-frequency mask generation model is a neural network which generates a time-frequency mask by using the sound feature vector **D22** and the text-embedded vector **D32** as inputs. For example, the neural network may have a plurality of convolution layers, activation functions, and pooling layers stacked with a skip connection interposed thereamong. In particular, when the sound feature vector **D22** is an amplitude spectrogram or a power spectrogram, the time-frequency mask generation unit **119a** may be, for example, conditional U-Net. That is, a sound feature vector which is an amplitude spectrogram or a power spectrogram is considered as an image, the image is input to a U-Net encoder implemented by stacking K convolution layers as an input, a U-Net feature map is calculated, the obtained U-Net feature map and the text-embedded vector **D32** are input to a U-Net decoder implemented by stacking K convolution layers, and a time-frequency mask which is an image having the same number of pixels as the sound feature vector **D22** is output. Each convolution layer $k=1, \dots, K$ of the U-Net encoder outputs a feature map V_k having different time-frequency resolutions from each other corresponding to that layer. The U-Net feature map is obtained by connecting the feature maps $V_1, V_2, \dots, V_K$ for all convolution layers. Each convolution layer $k=1, \dots, K$ of the U-Net decoder receives V_{K-k+1} in the U-Net feature map and the text-embedded vector **D32** as inputs. Alternatively, only the convolution layer $k=1$ may receive the feature map V_K and the text-embedded vector **D32** as the inputs, and other convolution layers $k \neq 1$ may receive only the feature map V_{K-k+1} as an input without the text-embedded vector **D32** as an input. If each convolution layer of the U-Net decoder receives the text-embedded vector **D32** as the input, there is an advantage that accuracy is increased if a training data set is abundant. When only the convolution layer $k=1$ receives the text-embedded vector **D32** as the input, there is an advantage that the number of weighting factors of the neural network is reduced to be small.

(57) The time-frequency mask multiplication unit **119b** multiplies the mixture-signal time-frequency domain representation **D21** by the time-frequency mask to generate an extraction-signal time-frequency domain representation **D41**.

(58) The learning unit **119c** minimizes a value of a loss function defined by a distance between the extraction-signal time-frequency domain representation **D41** and the target-signal time-frequency domain representation **D11** to learn parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model.

(59) The learning unit **119c** further calculates a convergence determination function used for determining whether convergence occurs. For example, the convergence determination function is defined by a magnitude of a change in the value of the loss function from a previous iteration (training). The convergence determination function may be defined by a magnitude of a change from a previous iteration (training) of a parameter of the feature extraction model, a magnitude of a change from a previous iteration (training) of a parameter of the text-embedded extraction model, a magnitude of a change from a previous iteration (training) of a parameter of the time-frequency mask generation model, or a product of these magnitudes of the changes. If the change is sufficiently small, it is determined that convergence occurs. For example, if the convergence determination function is smaller than a predetermined threshold value, it is determined that convergence occurs.

(60) (Outline of Operation)

(61) An outline of an operation of the learning subsystem **110** will be described. The learning subsystem **110** reads out, from the training data set database **160**, a set of three items of “the target-signal time waveform **D10**, the mixture-signal time waveform **D20** obtained by mixing the target-signal time waveform and a signal corresponding to noise other than the target signal, and the variable-length onomatopoeia text **D30**”.

(62) The target-signal time waveform **D10** is sequentially input to the target-signal frame division processing unit **111**, the target-signal window function multiplication unit **112**, and the target-signal frequency domain signal generation unit **113**, and is converted into the target-signal time-frequency domain representation **D11**.

(63) The mixture-signal time waveform **D20** is sequentially input to the mixture-signal frame division processing unit **114**, the mixture-signal window function multiplication unit **115**, and the mixture-signal frequency domain signal generation unit **116**, and is converted into the mixture-signal time-frequency domain representation **D21**.

(64) The mixture-signal time-frequency domain representation **D21** is input to the feature extraction unit **117**, and is converted into the sound feature vector **D22**.

(65) The variable-length onomatopoeia text **D30** is input to the phoneme conversion unit **118**, and is converted into the variable-length phoneme string **D31**. The variable-length phoneme string **D31** is input to the text-embedded extraction unit **119**, and is converted into the text-embedded vector **D32**.

(66) The sound feature vector **D22** and the text-embedded vector **D32** are input to the time-frequency mask generation unit **119a** to generate a time-frequency mask.

(67) The mixture-signal time-frequency domain representation **D21** and the time-frequency mask are multiplied in the time-frequency mask multiplication unit **119b** to generate the extraction-signal time-frequency domain representation **D41**. The extraction-signal time-frequency domain representation **D41** is input to the learning unit **119c**.

(68) The target-signal time-frequency domain representation **D11** and the extraction-signal time-frequency domain representation **D41** are input to the learning unit **119c**. The learning unit **119c** learns parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model, and updates the parameters. The feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model in which the parameters are updated are stored in the feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140**, respectively. The feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model in which the parameters are updated are also referred to as

“trained models” for convenience.

(69) (Specific Operation)

(70) A specific operation of the learning subsystem **110** will be described. FIG. **4** is an example of a processing flow of the learning subsystem **110**. The learning subsystem **110** executes the processing flow of FIG. **4**. When the learning subsystem **110** reads out, from the training data set database **160**, a set of three items including the target-signal time waveform **D10**, the mixture-signal time waveform **D20**, and the variable-length onomatopoeia text **D30**, the learning subsystem **110** starts processing from step **400** in FIG. **4** and proceeds to step **401**. When the learning subsystem **110** proceeds to step **401**, the learning subsystem **110** calculates, by the phoneme conversion unit **118**, the variable-length phoneme string **D31** from the variable-length onomatopoeia text **D30** by phoneme conversion processing (converts the onomatopoeia text **D30** into the phoneme string **D31**).

(71) Thereafter, the learning subsystem **110** proceeds to step **402**, and determines whether a learning end condition is satisfied. The learning end condition is a condition which is satisfied when either a condition 1 or a condition 2 described below is satisfied. The condition 1 is satisfied when a predetermined convergence condition is satisfied (for example, when the convergence determination function is smaller than a predetermined threshold value). The condition 2 is satisfied when a counter **C1** is larger than a threshold value **ThC** ($C1 > ThC$). The learning end condition may be only the condition 2.

(72) When the learning end condition is not satisfied, the learning subsystem **110** determines “NO” in step **402**, sequentially executes processing of steps **403** to **415** described below, and then returns to step **402**. Step **403**: The learning subsystem **110** calculates (extracts), by the text-embedded extraction unit **119**, the text-embedded vector **D32** from the phoneme string **D31** using a latest text-embedded extraction model. Step **404**: The learning subsystem **110** performs, by the mixture-signal frame division processing unit **114**, frame division of the mixture-signal time waveform **D20**, and calculates (outputs) a mixture-signal frame division signal. Step **405**: The learning subsystem **110** performs, by the mixture-signal window function multiplication unit **115**, window function multiplication, and converts the mixture-signal frame division signal into a mixture-signal window function multiplication signal. Step **406**: The learning subsystem **110** converts, by the mixture-signal frequency domain signal generation unit **116**, the mixture-signal window function multiplication signal into the mixture-signal time-frequency domain representation **D21**. Step **407**: The learning subsystem **110** calculates, by the feature extraction unit **117**, the sound feature vector **D22** from the mixture-signal time-frequency domain representation **D21**. In the present example, the learning subsystem **110** calculates, by the feature extraction unit **117**, the sound feature vector **D22** from the mixture-signal time-frequency domain representation **D21** using a latest feature extraction model. Step **408**: The learning subsystem **110** generates, by the time-frequency mask generation unit **119a**, the time-frequency mask from the sound feature vector **D22** and the text-embedded vector **D32** using a latest time-frequency mask generation model. Step **409**: The learning subsystem **110** multiplies, by the time-frequency mask multiplication unit **119b**, the mixture-signal time-frequency domain representation **D21** by the time-frequency mask to generate the extraction-signal time-frequency domain representation **D41**. Step **410**: The learning subsystem **110** performs, by the target-signal frame division processing unit **111**, frame division of the target-signal time waveform **D10**, and calculates (outputs) the target-signal frame division signal. Step **411**: The learning subsystem **110** performs, by the target-signal window function multiplication unit **112**, window function multiplication, and converts the target-signal frame division signal into the target-signal window function multiplication signal. Step **412**: The learning subsystem **110** performs, by the target-signal frequency domain signal generation unit **113**, short-time Fourier transform, and converts the target-signal window function multiplication signal into the target-signal time-frequency domain representation **D11**. Step **413**: The learning subsystem **110** minimizes, by the learning unit **119c**, the value of the loss function defined by the distance between the extraction-

signal time-frequency domain representation **D41** and the target-signal time-frequency domain representation **D11** to learn parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model (parameters of the neural networks (NNs)) (that is, update each model). Step **414**: The learning subsystem **110** calculates, by the learning unit **119c**, a convergence condition indicating whether convergence occurs. The convergence condition is defined by, for example, a magnitude of a change from a previous iteration (training) of the loss function. Alternatively, the convergence condition is defined by a magnitude of a change from a previous iteration (training) of the parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model. When the change is sufficiently small, it is determined that convergence occurs (step **402**). Step **415**: The learning subsystem **110** increases a current value of the counter **C1** by “1”. (73) When the learning end condition is satisfied in step **402**, the learning subsystem **110** determines “YES” in step **402** and proceeds to step **416** to store the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model (parameters of the neural networks (NNs)) in the respective databases (the feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140**). Thereafter, the learning subsystem **110** proceeds to step **495** and temporarily ends the processing flow.

(74) <Sound Extraction Subsystem>

(75) (Function of Sound Extraction Subsystem)

(76) A configuration of the sound extraction subsystem **150** will be described mainly for each function. FIG. 5 is a block diagram showing a configuration example of the sound extraction subsystem **150** for each function.

(77) As shown in FIG. 5, the sound extraction subsystem **150** includes a mixture-signal frame division processing unit **151**, a mixture-signal window function multiplication unit **152**, a mixture-signal frequency domain signal generation unit **153**, a feature extraction unit **154**, a time-frequency mask generation unit **155**, a phoneme conversion unit **156**, a text-embedded extraction unit **157**, a time-frequency mask multiplication unit **158**, and a phase restoration unit **159**. The mixture-signal frame division processing unit **151**, the mixture-signal window function multiplication unit **152**, the mixture-signal frequency domain signal generation unit **153**, the feature extraction unit **154**, the time-frequency mask generation unit **155**, the phoneme conversion unit **156**, the text-embedded extraction unit **157**, the time-frequency mask multiplication unit **158**, and the phase restoration unit **159** are implemented by various programs (not shown) stored in the ROM **202** and/or in the storage device **204** of the information processing device **200**.

(78) The mixture-signal frame division processing unit **151** performs frame division of a mixture-signal time waveform **D50**, and calculates (outputs) a mixture-signal frame division signal (not shown). The mixture-signal window function multiplication unit **152** performs window function multiplication and converts the mixture-signal frame division signal into a mixture-signal window function multiplication signal (not shown).

(79) The mixture-signal frequency domain signal generation unit **153** performs short-time Fourier transform, and converts the mixture-signal window function multiplication signal into a mixture-signal time-frequency domain representation **D51**. The frequency transform method such as the CQT may be used instead of the short-time Fourier transform, and the same processing as that of the learning subsystem **110** is performed.

(80) In the present example, similarly to the learning subsystem **110**, the feature extraction unit **154** converts the mixture-signal time-frequency domain representation **D51** into a sound feature vector using a latest feature extraction model (a feature extraction model which is a neural network in which the weighting factor parameter is variable). The mixture-signal time-frequency domain representation **D51** is input to the latest feature extraction model of the learning subsystem **110** updated immediately before, and a sound feature vector is calculated. When the learning subsystem

110 uses a logarithmic mel power spectrogram, a time series of the MFCC, a connection of delta or delta delta with respect to the logarithmic mel power spectrogram and the time series of the MFCC, or the like, the feature vector here may be a logarithmic mel power spectrogram, a time series of the MFCC, a connection of delta or delta-delta with respect to the logarithmic mel power spectrogram and the time series of the MFCC, or the like corresponding to that used in the learning subsystem **110**. In this case, the feature extraction unit **154** performs similar processing as the feature extraction unit **117** of the learning subsystem **110**.

(81) The phoneme conversion unit **156** outputs a variable-length phoneme string **D61** from a variable-length onomatopoeia text **D60** by phoneme conversion processing (converts the onomatopoeia text **D60** into the phoneme string **D61**).

(82) The text-embedded extraction unit **157** calculates (extracts), using a latest text-embedded extraction model, a text-embedded vector **D62** from the phoneme string **D61**.

(83) The time-frequency mask generation unit **155** generates, using a latest time-frequency mask generation model, a time-frequency mask from the sound feature vector **D52** and the text-embedded vector **D62**.

(84) The time-frequency mask multiplication unit **158** multiplies the mixture-signal time-frequency domain representation **D51** by the time-frequency mask to generate an extraction-signal time-frequency domain representation **D71**.

(85) The phase restoration unit **159** generates, using a known Griffin-Lim algorithm or the like, an extraction-signal time waveform **D72** from the extraction-signal time-frequency domain representation **D71**.

(86) (Outline of Operation)

(87) An outline of an operation of the sound extraction subsystem **150** will be described. As shown in FIG. 5, the mixture-signal time waveform **D50** is sequentially input to the mixture-signal frame division processing unit **151**, the mixture-signal window function multiplication unit **152**, and the mixture-signal frequency domain signal generation unit **153**, and is converted into the mixture-signal time-frequency domain representation **D51**. The mixture-signal time-frequency domain representation **D51** is input to the feature extraction unit **154**, and is converted into the sound feature vector **D52**.

(88) The variable-length onomatopoeia text **D60** is input to the phoneme conversion unit **156**, and is converted into the variable-length phoneme string **D61**. The variable-length phoneme string **D61** is input to the text-embedded extraction unit **157**, and is converted into the text-embedded vector **D62**. The sound feature vector **D52** and the text-embedded vector **D62** are input to the time-frequency mask generation unit **155** to generate the time-frequency mask.

(89) The mixture-signal time-frequency domain representation **D51** and the time-frequency mask are multiplied in the time-frequency mask multiplication unit **158** to generate the extraction-signal time-frequency domain representation **D71**. The extraction-signal time-frequency domain representation **D71** is input to the phase restoration unit **159** to generate the extraction-signal time waveform **D72**.

(90) (Specific Operation)

(91) A specific operation of the sound extraction subsystem **150** will be described. FIG. 6 is an example of a processing flow of the sound extraction subsystem **150**. The sound extraction subsystem **150** executes the processing flow of FIG. 6. When the mixture-signal time waveform **D50** and the variable-length onomatopoeia text **D60** are input, the sound extraction subsystem **150** starts processing from step **600** in FIG. 6, sequentially executes processing of steps **601** to **609** described below, and then proceeds to step **695** and temporarily ends the processing flow. Step **601**: The sound extraction subsystem **150** outputs, by the phoneme conversion unit **156**, the variable-length phoneme string **D61** from the variable-length onomatopoeia text **D60** by phoneme conversion processing (converts the onomatopoeia text **D60** into the phoneme string **D61**). Step **602**: The sound extraction subsystem **150** calculates (extracts), by the text-embedded extraction

unit **157**, the text-embedded vector **D62** from the phoneme string **D61** using a latest text-embedded extraction model. Step **603**: The sound extraction subsystem **150** performs, by the mixture-signal frame division processing unit **151**, frame division of the mixture-signal time waveform **D50**, and calculates (outputs) the mixture-signal frame division signal. Step **604**: The sound extraction subsystem **150** performs, by the mixture-signal window function multiplication unit **152**, window function multiplication, and converts the mixture-signal frame division signal into the mixture-signal window function multiplication signal. Step **605**: The sound extraction subsystem **150** performs, by the mixture-signal frequency domain signal generation unit **153**, short-time Fourier transform, and converts the mixture-signal window function multiplication signal into the mixture-signal time-frequency domain representation **D51**. Step **606**: The sound extraction subsystem **150** calculates, by the feature extraction unit **154**, the sound feature vector **D52** from the mixture-signal time-frequency domain representation **D51**. In the present example, the sound extraction subsystem **150** calculates, by the feature extraction unit **154**, the sound feature vector **D52** from the mixture-signal time-frequency domain representation **D51** using a latest feature extraction model. Step **607**: The sound extraction subsystem **150** generates, by the time-frequency mask generation unit **155**, the time-frequency mask from the sound feature vector **D52** and the text-embedded vector **D62** using a latest time-frequency mask generation model. Step **608**: The sound extraction subsystem **150** multiplies, by the time-frequency mask multiplication unit **158**, the mixture-signal time-frequency domain representation **D51** by the time-frequency mask to generate the extraction-signal time-frequency domain representation **D71**. Step **609**: The sound extraction subsystem **150** generates, by the phase restoration unit **159**, the extraction-signal time waveform **D72** from the extraction-signal time-frequency domain representation **D71** using the known Griffin-Lim algorithm or the like.

EXAMPLE

(92) FIG. 7 shows an example of extraction results of the sound extraction system **100**. A top row is power spectrograms of mixture signals. A horizontal axis represents time (seconds), and a vertical axis represents frequency (kHz). White indicates a time frequency having high power, and black indicates a time frequency having low power. In order to make it easy to understand an effect of the example, input mixture signals are signals obtained by mixing a plurality of sound events of the same type, and a range of a sound which the user wants to extract cannot be defined in advance as a certain type of event, so that extraction cannot be performed by an extraction method based on a type of event in the related art (that is, a related-art method corresponding to the related-art technique).

(93) A first column represents a task of extracting only a target signal corresponding to onomatopoeia “/poq/” from mixture signals obtained by mixing a plurality of metallic sounds. A second column represents a task of extracting only a target signal corresponding to onomatopoeia “/piririririN/” from mixture signals obtained by mixing a plurality of bell sounds. A third column represents a task of extracting only a target signal corresponding to onomatopoeia “/purururururu/” from mixture signals obtained by mixing a plurality of telephone rings. A fourth column represents a task of extracting only a target signal corresponding to onomatopoeia “/tichichichichichi/” from mixture signals obtained by mixing a plurality of hammering sounds. A fifth column represents a task of extracting only a target signal corresponding to onomatopoeia “/toqtututu/” from mixture signals obtained by mixing a plurality of sounds obtained by rolling a dice.

(94) In each column, a first row (“Mixture sound”) indicates input mixture signals, a second row (“Subclass-conditioned method”) indicates a result of an extraction method based on a type of event, a third row (“Onomatopoeia-conditioned method”) indicates a result of the extraction method of the present example, and a fourth row (“Ground truth”) indicates a target signal assumed to be correct. As compared with the mixture signals (first row) and the related-art method (second row), it can be seen that the extraction result (third row) of the present example is similar to the

correct target signal (fourth row). Therefore, the present example suggests an effect that a target sound can be extracted even when the range of the sound which the user wants to extract cannot be defined in advance as a certain type of event.

Effects

(95) As described above, the sound extraction system **100** according to the first embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. Further, the sound extraction system **100** according to the first embodiment can provide an infinite number of text, and can designate an infinite range of the sound. Therefore, the sound extraction system **100** according to the first embodiment can accurately extract a signal corresponding to a sound which the user wants to extract from mixture signals by providing text corresponding to the sound which the user wants to extract even when the range of the sound which the user wants to extract cannot be defined in advance as a certain type of event.

Second Embodiment

(96) A sound extraction system **800** according to a second embodiment of the invention will be described. FIG. **8** is a block configuration diagram showing a schematic configuration example of the sound extraction system **800** according to the second embodiment of the invention. As shown in FIG. **8**, the sound extraction system **800** is different from the sound extraction system **100** according to the first embodiment only in the following points. In the sound extraction system **800**, the learning subsystem **110** of the sound extraction system **100** according to the first embodiment is omitted, and a feature extraction model database **820**, a text-embedded extraction model database **830**, and a time-frequency mask generation model database **840** are used in which a feature extraction model, a text-embedded extraction model, and a time-frequency mask generation model trained in advance based on a general database of correspondence between an environmental sound and onomatopoeia are stored. This difference will be mainly described below.

(97) As shown in FIG. **8**, the sound extraction system **800** includes the sound extraction subsystem **150**, the feature extraction model database **820**, the text-embedded extraction model database **830**, the time-frequency mask generation model database **840**. When a mixture-signal time waveform and variable-length onomatopoeia text are input, the sound extraction subsystem **150** outputs an extraction-signal time waveform using an existing feature extraction model, text-embedded extraction model, and time-frequency mask generation model. Details of this processing are similar as those of the first embodiment except that the existing feature extraction model, text-embedded extraction model, and time-frequency mask generation model are used, and thus the description thereof will be omitted.

Effects

(98) As described above, similarly to the first embodiment, the sound extraction system **800** according to the second embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. Further, since the sound extraction system **800** according to the second embodiment can use the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model which are trained in advance based on the general database of the correspondence between the environmental sound and the onomatopoeia, new learning processing to be performed by the learning subsystem **110** as in the sound extraction system **100** according to the first embodiment is not required. There is an advantage that it is not required to construct a new training data set for each site.

Third Embodiment

(99) A sound extraction system **900** according to a third embodiment of the invention will be described. FIG. **9** is a block configuration diagram showing a schematic configuration example of the sound extraction system **900** according to the third embodiment of the invention. As shown in FIG. **9**, in the sound extraction system **900**, the learning subsystem **110** uses a feature extraction

model database **920**, a text-embedded extraction model database **930**, and a time-frequency mask generation model database **940** in which an existing feature extraction model, text-embedded extraction model, and time-frequency mask generation model trained in advance based on a general database of correspondence between an environmental sound and onomatopoeia are stored. The learning subsystem **110** uses a training data set for each site to learn, thereby optimizing a model according to the site and improving accuracy. The sound extraction system **900** according to the third embodiment is different from the sound extraction system **100** according to the first embodiment only in the above points. Therefore, this difference will be mainly described below.

(100) As shown in FIG. **9**, the sound extraction system **900** has a configuration in which the feature extraction model database **920**, the text-embedded extraction model database **930**, and the time-frequency mask generation model database **940** are added with respect to the sound extraction system **100** according to the first embodiment. Existing models stored in the feature extraction model database **920**, the text-embedded extraction model database **930**, and the time-frequency mask generation model database **940** are also referred to as an “initial feature extraction model, initial text-embedded extraction model, and initial time-frequency mask generation model” for convenience, and are also referred to as “initial trained models”.

(101) The learning subsystem **110** uses the training data set for each site to learn, thereby optimizing (updating) the models (the existing feature extraction model, text-embedded extraction model, and time-frequency mask generation model) according to the site, and storing the optimized models in the feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140**, respectively.

(102) When a mixture-signal time waveform and variable-length onomatopoeia text are input, the sound extraction subsystem **150** outputs an extraction-signal time waveform using the models obtained by optimizing the existing feature extraction model, text-embedded extraction model, and time-frequency mask generation model. Details of this processing are similar as those of the first embodiment except that the feature extraction model, text-embedded extraction model, and time-frequency mask generation model, which are obtained by optimizing the existing models, are used, and thus the description thereof will be omitted.

Effects

(103) As described above, similarly to the first embodiment, the sound extraction system **900** according to the third embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. Further, the sound extraction system **900** according to the third embodiment has an advantage of using an existing model so that a small number of training data sets are newly constructed for each site while improving accuracy of the model according to the site.

Fourth Embodiment

(104) A sound extraction system **1000** according to a fourth embodiment of the invention will be described. FIG. **10** is a block configuration diagram showing a schematic configuration example of the sound extraction system **1000** according to the fourth embodiment of the invention. As shown in FIG. **10**, the sound extraction system **1000** is different from the sound extraction system **100** according to the first embodiment only in that text such as descriptive text (for example, “clang-clang followed by boom”, or “impulsive sound followed by clang-clang”) is used as the text representing the range of the sound instead of the onomatopoeia. Therefore, this difference will be mainly described below.

(105) <Learning Subsystem>

(106) (Function of Learning Subsystem)

(107) FIG. **11** is a block diagram showing a configuration example of the learning subsystem **110** in the sound extraction system **1000** for each function. As shown in FIG. **11**, the learning subsystem **110** includes the target-signal frame division processing unit **111**, the target-signal window function multiplication unit **112**, the target-signal frequency domain signal generation unit **113**, the mixture-

signal frame division processing unit **114**, the mixture-signal window function multiplication unit **115**, the mixture-signal frequency domain signal generation unit **116**, the feature extraction unit **117**, the text-embedded extraction unit **119**, the time-frequency mask generation unit **119a**, the time-frequency mask multiplication unit **119b**, and the learning unit **119c**.

(108) (Outline of Operation)

(109) An outline of an operation of the learning subsystem **110** will be described. The learning subsystem **110** reads out, from the training data set database **160**, a set of three items including “the target-signal (signal corresponding to an extraction target sound) time waveform **D10**, the mixture-signal time waveform **D20** obtained by mixing the target-signal time waveform and a signal corresponding to noise other than the target signal (noise other than the extraction target sound), and a variable-length descriptive text **D1100** (descriptive text corresponding to the extraction target sound)”.

(110) The target-signal time waveform **D10** is sequentially input to the target-signal frame division processing unit **111**, the target-signal window function multiplication unit **112**, and the target-signal frequency domain signal generation unit **113**, and is converted into the target-signal time-frequency domain representation **D11**.

(111) The mixture-signal time waveform **D20** is sequentially input to the mixture-signal frame division processing unit **114**, the mixture-signal window function multiplication unit **115**, and the mixture-signal frequency domain signal generation unit **116**, and is converted into the mixture-signal time-frequency domain representation **D21**.

(112) The mixture-signal time-frequency domain representation **D21** is input to the feature extraction unit **117**, and is converted into the sound feature vector **D22**.

(113) The variable-length descriptive text **D1100** is input to the text-embedded extraction unit **119**, and is converted into an embedded vector **D1101** of the descriptive text **D1100** (text-embedded vector **D1101**).

(114) The sound feature vector **D22** and the text-embedded vector **D1101** are input to the time-frequency mask generation unit **119a** to generate a time-frequency mask.

(115) The mixture-signal time-frequency domain representation **D21** and the time-frequency mask are multiplied in the time-frequency mask multiplication unit **119b** to generate the extraction-signal time-frequency domain representation **D41**.

(116) When the target-signal time-frequency domain representation **D11** and the extraction-signal time-frequency domain representation **D41** are input to the learning unit **119c**, the learning unit **119c** learns parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model, and the parameters are updated.

(117) The feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model in which parameters are updated are stored in the feature extraction model database **120**, the text-embedded extraction model database **130**, and the time-frequency mask generation model database **140**, respectively.

(118) (Specific Operation)

(119) A specific operation of the learning subsystem **110** will be described. FIG. **12** is an example of a processing flow of the learning subsystem **110**. The learning subsystem **110** executes the processing flow of FIG. **12**. When the learning subsystem **110** reads out, from the training data set database **160**, a set of three items including the target-signal time waveform **D10**, the mixture-signal time waveform **D20**, and the variable-length descriptive text **D1100**, the learning subsystem **110** starts processing from step **1200** in FIG. **12**, proceeds to step **1201**, and determines whether the learning end condition is satisfied. The learning end condition is a condition which is satisfied when either a condition 1 or a condition 2 described below is satisfied. The condition 1 is satisfied when a predetermined convergence condition is satisfied (for example, when the convergence determination function is smaller than a predetermined threshold value). The condition 2 is satisfied when the counter **C1** is larger than the threshold value **ThC** ($C1 > ThC$). The learning end

condition may be only the condition 2.

(120) When the learning end condition is not satisfied, the learning subsystem **110** determines “NO” in step **1201**, sequentially executes processing of steps **1202** to **1214** described below, and then returns to step **1201**. Step **1202**: The learning subsystem **110** calculates (extracts), by the text-embedded extraction unit **119**, the text-embedded vector **D1101** from the variable-length descriptive text **D1100** using a latest text-embedded extraction model. Step **1203**: The learning subsystem **110** performs, by the mixture-signal frame division processing unit **114**, frame division of the mixture-signal time waveform, and calculates (outputs) a mixture-signal frame division signal. Step **1204**: The learning subsystem **110** performs, by the mixture-signal window function multiplication unit **115**, window function multiplication, and converts the mixture-signal frame division signal into a mixture-signal window function multiplication signal. Step **1205**: The learning subsystem **110** converts, by the mixture-signal frequency domain signal generation unit **116**, the mixture-signal window function multiplication signal into the mixture-signal time-frequency domain representation **D21**. Step **1206**: The learning subsystem **110** calculates, by the feature extraction unit **117**, the sound feature vector **D22** from the mixture-signal time-frequency domain representation **D21**. In the present example, the learning subsystem **110** calculates, by the feature extraction unit **117**, the sound feature vector **D22** from the mixture-signal time-frequency domain representation **D21** using a latest feature extraction model. Step **1207**: The learning subsystem **110** generates, by the time-frequency mask generation unit **119a**, a time-frequency mask from the sound feature vector **D22** and the text-embedded vector **D1101** using a latest time-frequency mask generation model. Step **1208**: The learning subsystem **110** multiplies, by the time-frequency mask multiplication unit **119b**, the mixture-signal time-frequency domain representation **D21** by the time-frequency mask to generate the extraction-signal time-frequency domain representation **D41**. Step **1209**: The learning subsystem **110** performs, by the target-signal frame division processing unit **111**, frame division of the target-signal time waveform **D10**, and calculates (outputs) a target-signal frame division signal. Step **1210**: The learning subsystem **110** performs, by the target-signal window function multiplication unit **112**, window function multiplication, and converts the target-signal frame division signal into a target-signal window function multiplication signal. Step **1211**: The learning subsystem **110** performs, by the target-signal frequency domain signal generation unit **113**, short-time Fourier transform, and converts the target-signal window function multiplication signal into the target-signal time-frequency domain representation **D11**. Step **1212**: The learning subsystem **110** minimizes, by the learning unit **119c**, a value of a loss function defined by a distance between the extraction-signal time-frequency domain representation **D41** and the target-signal time-frequency domain representation **D11** to learn (update) parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model (parameters of the neural networks (NNs)). Step **1213**: The learning subsystem **110** calculates a convergence condition indicating whether convergence occurs. The convergence condition is defined by, for example, a magnitude of a change from a previous iteration (training) of the loss function. Alternatively, the convergence condition is defined by a magnitude of a change from the previous iteration (training) of the parameters of the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model. When the change is sufficiently small, it is determined that convergence occurs (step **1201**). Step **1214**: The learning subsystem **110** increases a current value of the counter **C1** by “1”.

(121) When the learning end condition is satisfied in step **1201**, the learning subsystem **110** determines “YES” in step **1201** and proceeds to step **1215** to store the feature extraction model, the text-embedded extraction model, and the time-frequency mask generation model (parameters of the neural networks (NNs)) in the respective databases. Thereafter, the learning subsystem **110** proceeds to step **1295** and temporarily ends the processing flow.

(122) <Sound Extraction Subsystem>

(123) (Function of Sound Extraction Subsystem)

(124) Hereinafter, a configuration of the sound extraction subsystem **150** will be described mainly for each function. FIG. **13** is a block diagram showing a configuration example of the sound extraction subsystem **150** for each function.

(125) As shown in FIG. **13**, the sound extraction subsystem **150** includes the mixture-signal frame division processing unit **151**, the mixture-signal window function multiplication unit **152**, the mixture-signal frequency domain signal generation unit **153**, the feature extraction unit **154**, the time-frequency mask generation unit **155**, the text-embedded extraction unit **157**, the time-frequency mask multiplication unit **158**, and the phase restoration unit **159**.

(126) (Outline of Operation)

(127) As shown in FIG. **13**, the mixture-signal time waveform **D50** is sequentially input to the mixture-signal frame division processing unit **151**, the mixture-signal window function multiplication unit **152**, and the mixture-signal frequency domain signal generation unit **153**, and is converted into the mixture-signal time-frequency domain representation **D51**. The mixture-signal time-frequency domain representation **D51** is input to the feature extraction unit **154**, and is converted into the sound feature vector **D52**.

(128) A variable-length descriptive text **D1300** is input to the text-embedded extraction unit **157**, and is converted into an embedded vector **D1301** of the descriptive text **D1300** (text-embedded vector **D1301**). The sound feature vector **D52** and the text-embedded vector **D1301** are input to the time-frequency mask generation unit **155** to generate a time-frequency mask.

(129) The mixture-signal time-frequency domain representation **D51** and the time-frequency mask are multiplied in the time-frequency mask multiplication unit **158** to generate the extraction-signal time-frequency domain representation **D71**. The extraction-signal time-frequency domain representation **D71** is input to the phase restoration unit **159** to generate the extraction-signal time waveform **D72**.

(130) (Specific Operation)

(131) FIG. **14** is an example of a processing flow of the sound extraction subsystem **150**. The sound extraction subsystem **150** executes the processing flow of FIG. **14**. When the mixture-signal time waveform **D50** and the variable-length descriptive text **D1300** are input, the sound extraction subsystem **150** starts processing from step **1400** in FIG. **14**, sequentially executes processing of steps **1401** to **1408** described below, and then proceeds to step **1495** and temporarily ends the processing flow. Step **1401**: The sound extraction subsystem **150** calculates (extracts), by the text-embedded extraction unit **157**, the text-embedded vector **D1301** from the variable-length descriptive text **D1300** using a latest text-embedded extraction model. Step **1402**: The sound extraction subsystem **150** performs, by the mixture-signal frame division processing unit **151**, frame division of the mixture-signal time waveform, and calculates (outputs) the mixture-signal frame division signal. Step **1403**: The sound extraction subsystem **150** performs, by the mixture-signal window function multiplication unit **152**, window function multiplication, and converts the mixture-signal frame division signal into the mixture-signal window function multiplication signal. Step **1404**: The sound extraction subsystem **150** performs, by the mixture-signal frequency domain signal generation unit **153**, short-time Fourier transform, and converts the mixture-signal window function multiplication signal into the mixture-signal time-frequency domain representation **D51**. Step **1405**: The sound extraction subsystem **150** calculates, by the feature extraction unit **154**, the sound feature vector **D52** from the mixture-signal time-frequency domain representation **D51**. In the present example, the sound extraction subsystem **150** calculates, by the feature extraction unit **154**, the sound feature vector **D52** from the mixture-signal time-frequency domain representation **D51** using a latest feature extraction model. Step **1406**: The sound extraction subsystem **150** generates, by the time-frequency mask generation unit **155**, the time-frequency mask from the sound feature vector **D52** and the text-embedded vector **D1301** using a latest time-frequency mask generation model. Step **1407**: The sound extraction subsystem **150** multiplies, by the time-frequency mask multiplication unit **158**, the mixture-signal time-frequency domain representation

D51 by the time-frequency mask to generate the extraction-signal time-frequency domain representation D71. Step **1408**: The sound extraction subsystem **150** generates, using the known Griffin-Lim algorithm or the like, the extraction-signal time waveform D72 from the extraction-signal time-frequency domain representation D71.

Effects

(132) As described above, the sound extraction system **1000** according to the fourth embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. With such a basic configuration, even when a range of the sound which the user wants to extract cannot be defined in advance as a certain type of event, the sound extraction system **1000** according to the fourth embodiment can extract the sound. The descriptive text is relatively generic and can be used across the application site.

Fifth Embodiment

(133) A sound extraction system **1500** according to a fifth embodiment of the invention will be described. FIG. **15** is a block configuration diagram showing a schematic configuration example of the sound extraction system **1500** according to the fifth embodiment of the invention. As shown in FIG. **15**, the sound extraction system **1500** includes the learning subsystem **110**, a signal extraction model database **1510**, the text-embedded extraction model database **130**, the sound extraction subsystem **150**, and the training data set database **160**. Differences from the first embodiment shown in FIG. **1** will be described. The learning subsystem **110** executes learning processing, outputs a signal extraction model and a text-embedded extraction model, and stores the signal extraction model and the text-embedded extraction model in respective databases. That is, the learning subsystem **110** stores the signal extraction model in the signal extraction model database **1510**, and stores the text-embedded extraction model in the text-embedded extraction model database **130**.

(134) The sound extraction subsystem **150** reads out the signal extraction model and the text-embedded extraction model from the databases (the signal extraction model database **1510** and the text-embedded extraction model database **130**), and executes sound extraction processing based on (using) the read models. Accordingly, the sound extraction subsystem **150** extracts an extraction-signal time waveform from the mixture-signal time waveform and the variable-length onomatopoeia text. Further, the sound extraction subsystem **150** outputs the extraction-signal time waveform.

(135) <Learning Subsystem>

(136) (Function of Learning Subsystem)

(137) Hereinafter, a configuration of the learning subsystem **110** will be described mainly for each function. FIG. **16** is a block diagram showing a configuration example of the learning subsystem **110** for each function. As shown in FIG. **16**, the learning subsystem **110** includes the phoneme conversion unit **118**, the text-embedded extraction unit **119**, a signal extraction unit **1600**, and the learning unit **119c**. The phoneme conversion unit **118**, the text-embedded extraction unit **119**, the signal extraction unit **1600**, and the learning unit **119c** are implemented by various programs (not shown) stored in the ROM **202** and/or the storage device **204** of the information processing device **200**.

(138) The signal extraction unit **1600** generates, using a latest signal extraction model, an extraction-signal time waveform D**1600** from the mixture-signal time waveform D**20** and the text-embedded vector D**32**.

(139) The signal extraction model is a neural network which receives the mixture-signal time waveform D**20** and the text-embedded vector D**32** and outputs the extraction-signal time waveform D**1600**. The signal extraction model may be, for example, a neural network including only fully connected layers, or may be a neural network in which a plurality of convolution layers, activation functions, and pooling layers are stacked with a self-attention layer or a skip connection interposed thereamong. When a time-frequency mask is used as in the first embodiment, a time-frequency

representation is required. However, the time-frequency representation is not necessarily appropriate in terms of extraction accuracy. On the other hand, for the signal extraction model here, the time waveform is directly inputted to the neural network, so that there is an advantage that a representation with high extraction accuracy can be obtained if a training data set is sufficiently large.

(140) In the case of the neural network including only fully connected layers, there is an advantage that the extraction accuracy is high when a scale of the training data set is large, and in the case of the neural network in which a plurality of convolution layers, activation functions, and pooling layers are stacked with a self-attention layer or a skip connection interposed thereamong, there is an advantage that the extraction accuracy is high even when the training data set is small.

(141) The signal extraction model, as well-known Conv-TasNet, may be a model including: an encoder which receives the mixture-signal time waveform **D20** and outputs a feature vector time series; a time feature mask generation neural network which receives the feature vector time series and the embedded vector **D32** and calculates a two-dimensional mask (time feature mask) on a time axis and a feature axis; a multiplication mechanism which multiplies the time feature mask by the feature vector time series and calculates an extracted feature vector time series; and a decoder which receives the extracted feature vector time series and generates the extraction-signal time waveform **D1600**. Each of the encoder and the decoder is, for example, a neural network including a one-dimensional convolution layer. The time feature mask generation neural network may be a neural network including only fully connected layers, or may be a neural network in which a plurality of convolution layers, activation functions, and pooling layers are stacked with a self-attention layer or a skip connection interposed thereamong. When the time-frequency mask is used as in the first embodiment, the time-frequency representation is required, but high extraction accuracy cannot be necessarily expected when the time-frequency representation is used. On the other hand, in the signal extraction model which performs conversion into a time feature inside the neural network, since a time feature representation trained to increase the extraction accuracy is used, there is an advantage that the extraction accuracy is higher than that when the time-frequency representation is used.

(142) The learning unit **119c** minimizes a value of a loss function defined by a distance between the extraction-signal time waveform **D1600** and the target-signal time waveform **D10** to learn parameters of the signal extraction model and the text-embedded extraction model.

(143) The learning unit **119c** further calculates a convergence determination function used for determining whether convergence occurs. For example, the convergence determination function is defined by a magnitude of a change in the value of the loss function from a previous iteration (training). The convergence determination function may be defined by a magnitude of a change from a previous iteration (training) of a parameter of the signal extraction model, a magnitude of a change from a previous iteration (training) of a parameter of the text-embedded extraction model, or a product of these magnitudes of the changes. If the change is sufficiently small, it is determined that convergence occurs. For example, if the convergence determination function is smaller than a predetermined threshold value, it is determined that convergence occurs.

(144) (Outline of Operation)

(145) An outline of an operation of the learning subsystem **110** will be described. The learning subsystem **110** reads out, from the training data set database **160**, a set of three items including “the target-signal time waveform **D10**, the mixture-signal time waveform **D20** obtained by mixing the target-signal time waveform and a signal corresponding to noise other than the target signal, and the variable-length onomatopoeia text **D30**”.

(146) The variable-length onomatopoeia text **D30** is input to the phoneme conversion unit **118**, and is converted into the variable-length phoneme string **D31**. The variable-length phoneme string **D31** is input to the text-embedded extraction unit **119**, and is converted into the text-embedded vector **D32**.

(147) The mixture-signal time waveform D20 and the text-embedded vector D32 are input to the signal extraction unit **1600** to generate the extraction-signal time waveform D1600.

(148) The target-signal time waveform D10 and the extraction-signal time waveform D1600 are input to the learning unit **119c**. The learning unit **119c** learns parameters of the signal extraction model and the text-embedded extraction model, and updates the parameters. The signal extraction model and the text-embedded extraction model in which the parameters are updated are stored in the signal extraction model database **1510** and the text-embedded extraction model database **130**, respectively. The signal extraction model and the text-embedded extraction model in which the parameters are updated are also referred to as “trained models” for convenience.

(149) (Specific Operation)

(150) A specific operation of the learning subsystem **110** will be described. FIG. **17** is an example of a processing flow of the learning subsystem **110**. The learning subsystem **110** executes the processing flow of FIG. **17**. When the learning subsystem **110** reads out, from the training data set database **160**, a set of three items including the target-signal time waveform D10, the mixture-signal time waveform D20, and the variable-length onomatopoeia text D30, the learning subsystem **110** starts processing from step **1700** in FIG. **17** and proceeds to step **1701**. When the learning subsystem **110** proceeds to step **1701**, the learning subsystem **110** calculates, by the phoneme conversion unit **118**, the variable-length phoneme string D31 from the variable-length onomatopoeia text D30 by phoneme conversion processing (converts the onomatopoeia text D30 into the phoneme string D31).

(151) Thereafter, the learning subsystem **110** proceeds to step **1702**, and determines whether a learning end condition is satisfied. The learning end condition is a condition which is satisfied when either a condition 1 or a condition 2 described below is satisfied. The condition 1 is satisfied when a predetermined convergence condition is satisfied (for example, when the convergence determination function is smaller than a predetermined threshold value). The condition 2 is satisfied when the counter C1 is larger than the threshold value ThC ($C1 > ThC$). The learning end condition may be only the condition 2.

(152) When the learning end condition is not satisfied, the learning subsystem **110** determines “NO” in step **1702**, sequentially executes processing of steps **1703** to **1707** described below, and then returns to step **1702**. Step **1703**: The learning subsystem **110** calculates (extracts), by the text-embedded extraction unit **119**, the text-embedded vector D32 from the phoneme string D31 using a latest text-embedded extraction model. Step **1704**: The learning subsystem **110** generates, by the signal extraction unit **1600**, the extraction-signal time waveform D1600 from the mixture-signal time waveform D20 and the text-embedded vector D32 using a latest signal extraction model. Step **1705**: The learning subsystem **110** minimizes, by the learning unit **119c**, a value of a loss function defined by a distance between the extraction-signal time waveform D1600 and the target-signal time waveform D10 to learn parameters of the signal extraction model and the text-embedded extraction model (parameters of the neural networks (NNs)) (that is, update each model). Step **1706**: The learning subsystem **110** calculates, by the learning unit **119c**, a convergence condition indicating whether convergence occurs. The convergence condition is defined by, for example, a magnitude of a change from the previous iteration (training) of the loss function. Alternatively, the convergence condition is defined by a magnitude of a change from a previous iteration (training) of the parameters of the signal extraction model and the text-embedded extraction model. If the change is sufficiently small, it is determined that convergence occurs (step **1702**). Step **1707**: The learning subsystem **110** increases a current value of the counter C1 by “1”.

(153) When the learning end condition is satisfied in step **1702**, the learning subsystem **110** determines “YES” in step **1702** and proceeds to step **1708** to store the signal extraction model and the text-embedded extraction model (parameters of the neural networks (NNs)) in the respective databases (the signal extraction model database **1510** and the text-embedded extraction model database **130**). Thereafter, the learning subsystem **110** proceeds to step **1795** and temporarily ends

the processing flow.

(154) <Sound Extraction Subsystem>

(155) (Function of Sound Extraction Subsystem)

(156) A configuration of the sound extraction subsystem **150** will be described mainly for each function. FIG. **18** is a block diagram showing a configuration example of the sound extraction subsystem **150** for each function.

(157) As shown in FIG. **18**, the sound extraction subsystem **150** includes the phoneme conversion unit **156**, the text-embedded extraction unit **157**, and a signal extraction unit **1800**. The phoneme conversion unit **156**, the text-embedded extraction unit **157**, and the signal extraction unit **1800** are implemented by various programs (not shown) stored in the ROM **202** and/or the storage device **204** of the information processing device **200**.

(158) The phoneme conversion unit **156** outputs the variable-length phoneme string **D61** from the variable-length onomatopoeia text **D60** by phoneme conversion processing (converts the onomatopoeia text **D60** into the phoneme string **D61**).

(159) The text-embedded extraction unit **157** calculates (extracts), using a latest text-embedded extraction model, the text-embedded vector **D62** from the phoneme string **D61**.

(160) The signal extraction unit **1800** generates, using a latest signal extraction model, the extraction-signal time waveform **D72** from the mixture-signal time waveform **D50** and the text-embedded vector **D62**.

(161) (Specific Operation)

(162) A specific operation of the sound extraction subsystem **150** will be described. FIG. **19** is an example of a processing flow of the sound extraction subsystem **150**. The sound extraction subsystem **150** executes the processing flow of FIG. **19**. When the mixture-signal time waveform **D50** and the variable-length onomatopoeia text **D60** are input, the sound extraction subsystem **150** starts processing from step **1900** in FIG. **19**, sequentially executes processing of steps **1901** to **1903** described below, and then proceeds to step **1995** and temporarily ends the processing flow. Step **1901**: The sound extraction subsystem **150** outputs, by the phoneme conversion unit **156**, the variable-length phoneme string **D61** from the variable-length onomatopoeia text **D60** by phoneme conversion processing (converts the onomatopoeia text **D60** into the phoneme string **D61**). Step **1902**: The sound extraction subsystem **150** calculates (extracts), by the text-embedded extraction unit **157**, the text-embedded vector **D62** from the phoneme string **D61** using a latest text-embedded extraction model. Step **1903**: The sound extraction subsystem **150** generates, by the signal extraction unit **1800**, the extraction-signal time waveform **D72** from the mixture-signal time waveform **D50** and the text-embedded vector **D62** using a latest signal extraction model.

Effects

(163) As described above, the sound extraction system **1500** according to the fifth embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. Further, the sound extraction system **1500** according to the fifth embodiment can provide an infinite number of text, and can designate an infinite range of the sound. Therefore, the sound extraction system **1500** according to the fifth embodiment can accurately extract, from mixture signals, a signal corresponding to a sound which the user wants to extract by providing text corresponding to the sound which the user wants to extract even when the range of the sound which the user wants to extract cannot be defined in advance as a certain type of event. Further, unlike the first embodiment, in the sound extraction system **1500** according to the fifth embodiment, the mixture-signal time waveform **D50** is directly inputted to the neural network without being converted into the time-frequency representation, and thus a reduction in the extraction accuracy due to use of the time-frequency representation can be avoided. Since the sound extraction system **1500** according to the fifth embodiment generates the extraction-signal time waveform **D72** without performing the phase restoration processing, there is also an advantage that distortion due to the phase restoration processing does not occur.

Sixth Embodiment

(164) A sound extraction system **2000** according to a sixth embodiment of the invention will be described. FIG. **20** is a block configuration diagram showing a schematic configuration example of the sound extraction system **2000** according to the sixth embodiment of the invention. As shown in FIG. **20**, the sound extraction system **2000** is different from the sound extraction system **1500** according to the fifth embodiment only in the following points.

(165) In the sound extraction system **2000**, the learning subsystem **110** of the sound extraction system **1500** according to the fifth embodiment is omitted, and a signal extraction model database **2010** and the text-embedded extraction model database **830** are used in which a signal extraction model and a text-embedded extraction model trained in advance based on a general database of correspondence between an environmental sound and onomatopoeia are stored. This difference will be mainly described below.

(166) As shown in FIG. **20**, the sound extraction system **2000** includes the sound extraction subsystem **150**, the signal extraction model database **2010**, and the text-embedded extraction model database **830**. When a mixture-signal time waveform and variable-length onomatopoeia text are input, the sound extraction subsystem **150** outputs an extraction-signal time waveform using an existing signal extraction model and text-embedded extraction model. Details of this processing are similar as those of the fifth embodiment except that the existing signal extraction model and text-embedded extraction model are used, and thus the description thereof will be omitted.

Effects

(167) As described above, similarly to the fifth embodiment, the sound extraction system **2000** according to the sixth embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. In the sound extraction system **2000** according to the sixth embodiment, the mixture-signal time waveform is directly inputted to the neural network without being converted into the time-frequency representation, and thus a reduction in the extraction accuracy due to the use of the time-frequency representation can be avoided. Since the sound extraction system **2000** according to the sixth embodiment generates the extraction-signal time waveform without performing the phase restoration processing, there is also an advantage that the distortion due to the phase restoration processing does not occur. Further, since the sound extraction system **2000** according to the sixth embodiment can use the signal extraction model and the text-embedded extraction model which are trained in advance based on the general database of the correspondence between the environmental sound and the onomatopoeia, new learning processing to be performed by the learning subsystem **110** as in the sound extraction system **1500** according to the fifth embodiment is not required. There is an advantage that it is not required to construct a new training data set for each site.

Seventh Embodiment

(168) A sound extraction system **2100** according to a seventh embodiment of the invention will be described. FIG. **21** is a block configuration diagram showing a schematic configuration example of the sound extraction system **2100** according to the seventh embodiment of the invention. As shown in FIG. **21**, in the sound extraction system **2100**, the learning subsystem **110** uses a signal extraction model database **2110** and the text-embedded extraction model database **930** in which an existing signal extraction model and text-embedded extraction model trained in advance based on a general database of correspondence between an environmental sound and the onomatopoeia are stored. The learning subsystem **110** uses a training data set for each site to learn, thereby optimizing a model according to the site and improving accuracy. The sound extraction system **2100** according to the seventh embodiment is different from the sound extraction system **1500** according to the fifth embodiment only in the above points. Therefore, this difference will be mainly described below.

(169) As shown in FIG. **21**, the sound extraction system **2100** has a configuration in which the signal extraction model database **2110** and the text-embedded extraction model database **930** are added with respect to the sound extraction system **1500** according to the fifth embodiment. Existing

models stored in the signal extraction model database **2110** and the text-embedded extraction model database **930** are also referred to as an “initial signal extraction model and initial text-embedded extraction model” for convenience, and are also referred to as “initial trained models”. (170) The learning subsystem **110** uses the training data set for each site to learn, thereby optimizing (updating) the models (the existing signal extraction model and text-embedded extraction model) according to the site, and storing the optimized models in the signal extraction model database **1510** and the text-embedded extraction model database **130**, respectively.

(171) When a mixture-signal time waveform and variable-length onomatopoeia text are input, the sound extraction subsystem **150** outputs an extraction-signal time waveform by optimizing the existing signal extraction model and text-embedded extraction model. Details of this processing are similar as those of the fifth embodiment except that the signal extraction model and the text-embedded extraction model obtained by optimizing the existing model are used, and thus the description thereof will be omitted.

Effects

(172) As described above, similarly to the fifth embodiment, the sound extraction system **2100** according to the seventh embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. In the sound extraction system **2100** according to the seventh embodiment, the mixture-signal time waveform is directly inputted to the neural network without being converted into the time-frequency representation, and thus a reduction in the extraction accuracy due to the use of the time-frequency representation can be avoided. Since the sound extraction system **2100** according to the seventh embodiment generates the extraction-signal time waveform without performing the phase restoration processing, there is also an advantage that the distortion due to the phase restoration processing does not occur. Further, the sound extraction system **2100** according to the seventh embodiment has an advantage of using an existing model so that a small number of training data sets are newly constructed for each site while improving accuracy of the model according to the site.

Eighth Embodiment

(173) A sound extraction system **2200** according to an eighth embodiment of the invention will be described. FIG. **22** is a block configuration diagram showing a schematic configuration example of the sound extraction system **2200** according to the eighth embodiment of the invention. As shown in FIG. **22**, the sound extraction system **2200** is different from the sound extraction system **1500** according to the fifth embodiment only in that text such as descriptive text (for example, “clang-clang followed by boom”, or “impulsive sound followed by clang-clang”) is used as text representing a range of a sound instead of the onomatopoeia. Therefore, this difference will be mainly described below.

(174) <Learning Subsystem>

(175) (Function of Learning Subsystem)

(176) FIG. **23** is a block diagram showing a configuration example of the learning subsystem **110** in the sound extraction system **2200** for each function. As shown in FIG. **23**, the learning subsystem **110** includes the text-embedded extraction unit **119**, the signal extraction unit **1600**, and the learning unit **119c**.

(177) (Outline of Operation)

(178) An outline of an operation of the learning subsystem **110** will be described. The learning subsystem **110** reads out, from the training data set database **160**, a set of three items including “the target-signal (signal corresponding to an extraction target sound) time waveform **D10**, the mixture-signal time waveform **D20** obtained by mixing the target-signal time waveform and a signal corresponding to noise other than the target signal (noise other than the extraction target sound), and a variable-length descriptive text **D1100** (descriptive text corresponding to the extraction target sound)”.

(179) The variable-length descriptive text **D1100** is input to the text-embedded extraction unit **119**, and is converted into an embedded vector **D1101** of the descriptive text **D1100** (text-embedded vector **D1101**).

(180) The mixture-signal time waveform **D20** and the text-embedded vector **D1101** are input to the signal extraction unit **1600** to generate the extraction-signal time waveform **D1600**.

(181) When the target-signal time waveform **D10** and the extraction-signal time waveform **D1600** are input to the learning unit **119c**, the learning unit **119c** learns parameters of the signal extraction model and the text-embedded extraction model, and the parameters are updated.

(182) The signal extraction model and the text-embedded extraction model in which the parameters are updated are stored in the signal extraction model database **1510** and the text-embedded extraction model database **130**, respectively.

(183) (Specific Operation)

(184) A specific operation of the learning subsystem **110** will be described. FIG. **24** is an example of a processing flow of the learning subsystem **110**. The learning subsystem **110** executes the processing flow of FIG. **24**. When the learning subsystem **110** reads out, from the training data set database **160**, a set of three items including the target-signal time waveform **D10**, the mixture-signal time waveform **D20**, and the variable-length descriptive text **D1100**, the learning subsystem **110** starts processing from step **2400** in FIG. **24**, proceeds to step **2401**, and determines whether the learning end condition is satisfied. The learning end condition is a condition which is satisfied when either a condition 1 or a condition 2 described below is satisfied. The condition 1 is satisfied when a predetermined convergence condition is satisfied (for example, when the convergence determination function is smaller than a predetermined threshold value). The condition 2 is satisfied when the counter **C1** is larger than the threshold value **ThC** ($C1 > ThC$). The learning end condition may be only the condition 2.

(185) When the learning end condition is not satisfied, the learning subsystem **110** determines “NO” in step **2401**, sequentially executes processing of steps **2402** to **2406** described below, and then returns to step **2401**. Step **2402**: The learning subsystem **110** calculates (extracts), by the text-embedded extraction unit **119**, the text-embedded vector **D1101** from the variable-length descriptive text **D1100** using a latest text-embedded extraction model. Step **2403**: The learning subsystem **110** generates, by the signal extraction unit **1600**, the extraction-signal time waveform **D1600** from the mixture-signal time waveform **D20** and the text-embedded vector **D1101** using a latest signal extraction model. Step **2404**: The learning subsystem **110** minimizes, by the learning unit **119c**, a value of a loss function defined by a distance between the extraction-signal time waveform **D1600** and the target-signal time waveform **D10** to learn (update) parameters of the signal extraction model and the text-embedded extraction model (parameters of the neural networks (NNs)). Step **2405**: The learning subsystem **110** calculates a convergence condition indicating whether convergence occurs. The convergence condition is defined by, for example, a magnitude of a change from a previous iteration (training) of the loss function. Alternatively, the convergence condition is defined by a magnitude of a change from the previous iteration (training) of the parameters of the signal extraction model and the text-embedded extraction model. If the change is sufficiently small, it is determined that convergence occurs (step **2401**). Step **2406**: The learning subsystem **110** increases a current value of the counter **C1** by “1”.

(186) When the learning end condition is satisfied in step **2401**, the learning subsystem **110** determines “YES” in step **2401** and proceeds to step **2407** to store the signal extraction model and the text-embedded extraction model (parameters of the neural networks (NNs)) in the respective databases. Thereafter, the learning subsystem **110** proceeds to step **2495** and temporarily ends the processing flow.

(187) <Sound Extraction Subsystem>

(188) (Function of Sound Extraction Subsystem)

(189) Hereinafter, a configuration of the sound extraction subsystem **150** will be described mainly

for each function. FIG. 25 is a block diagram showing a configuration example of the sound extraction subsystem **150** for each function.

(190) As shown in FIG. 25, the sound extraction subsystem **150** includes the text-embedded extraction unit **157** and the signal extraction unit **1800**.

(191) (Outline of Operation)

(192) As shown in FIG. 25, the variable-length descriptive text **D1300** is input to the text-embedded extraction unit **157**, and is converted into the embedded vector **D1301** of the descriptive text **D1300** (text-embedded vector **D1301**). The mixture-signal time waveform **D50** and the text-embedded vector **D1301** are input to the signal extraction unit **1800** to generate the extraction-signal time waveform **D72**.

(193) (Specific Operation)

(194) FIG. 26 is an example of a processing flow of the sound extraction subsystem **150**. The sound extraction subsystem **150** executes the processing flow of FIG. 26. When the mixture-signal time waveform **D50** and the variable-length descriptive text **D1300** are input, the sound extraction subsystem **150** starts processing from step **2600** in FIG. 26, sequentially executes processing of steps **2601** and **2602** described below, and then proceeds to step **2695** and temporarily ends the processing flow. Step **2601**: The sound extraction subsystem **150** calculates (extracts), by the text-embedded extraction unit **157**, the text-embedded vector **D1301** from the variable-length descriptive text **D1300** using a latest text-embedded extraction model. Step **2602**: The sound extraction subsystem **150** generates, by the signal extraction unit **1800**, the extraction-signal time waveform **D72** from the mixture-signal time waveform **D50** and the text-embedded vector **D1301** using a latest signal extraction model.

Effects

(195) As described above, the sound extraction system **2200** according to the eighth embodiment of the invention can accurately extract (extract or emphasize), from mixture signals, a signal corresponding to a sound which the user wants to extract. With such a basic configuration, even when a range of the sound which the user wants to extract cannot be defined in advance as a certain type of event, the sound extraction system **2200** according to the eighth embodiment can extract the sound. The descriptive text is relatively generic and can be used across the application site. In the sound extraction system **2200** according to the eighth embodiment, the mixture-signal time waveform is directly inputted to the neural network without being converted into the time-frequency representation, and thus a reduction in the extraction accuracy due to the use of the time-frequency representation can be avoided. Since the sound extraction system **2200** according to the eighth embodiment generates the extraction-signal time waveform without performing the phase restoration processing, there is also an advantage that distortion due to the phase restoration processing does not occur.

Modifications

(196) The invention is not limited to the above-mentioned embodiments, and various modifications can be made within the scope of the invention. Further, the embodiments described above can be combined with each other without departing from the scope of the invention. Further, within the scope of the invention, a part of a configuration of an embodiment can be replaced with a configuration of another embodiment. Further, within the scope of the invention, it is also possible to add a configuration of another embodiment to a configuration of an embodiment. Further, within the scope of the invention, a part of the configuration of each embodiment can be added to, deleted from, or replaced with another configuration.

(197) Further, in each of the above embodiments, text input to the sound extraction subsystem **150** may be input by operating an operation device such as a keyboard. Further, in each of the above embodiments, the text input to the sound extraction subsystem **150** may be input by converting human speech into text by speech recognition technology. Further, in each of the above

embodiments, mixture signals input to the sound extraction subsystem **150** may be input from an acoustic device such as a microphone.

Claims

1. A sound extraction system comprising: a sound extraction device configured to extract, from mixture signals including a signal corresponding to an extraction target sound, the signal corresponding to the extraction target sound, wherein the sound extraction device is configured to: extract the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and a text representing a range of the extraction target sound, generate, based on the mixture signals and the text representing the range of the extraction target sound, a time-frequency mask for extracting the signal corresponding to the extraction target sound, and apply the time-frequency mask to the mixture signals and extract the signal corresponding to the extraction target sound from the mixture signals; and a learning device configured to generate a trained model, which is used for generating the time-frequency mask, by executing machine learning using a training data set including a target signal corresponding to the extraction target sound, training mixture signals obtained by mixing the target signal and a signal corresponding to noise other than the extraction target sound, and a training text corresponding to the extraction target sound, wherein the sound extraction device is also configured to: generate, using the trained model generated by the learning device, the time-frequency mask based on the mixture signals and the text representing the range of the extraction target sound, wherein the learning device is configured to generate, as the trained model, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a time-frequency mask generation model configured to generate the time-frequency mask from a sound feature vector of the mixture signals and the text-embedded vector, and the sound extraction device is configured to: calculate the sound feature vector of the mixture signals from the mixture signals, calculate, using the text-embedded extraction model generated by the learning device, the text-embedded vector from the text representing the range of the extraction target sound, and generate, using the time-frequency mask generation model generated by the learning device, the time-frequency mask from the calculated text-embedded vector and sound feature vector of the mixture signals.
2. The sound extraction system according to claim 1, further comprising: a storage device which stores a trained model used for generating the time-frequency mask, wherein the sound extraction device is configured to generate, using the trained model, the time-frequency mask based on the mixture signals and the text representing the range of the extraction target sound.
3. The sound extraction system according to claim 2, wherein the storage device stores, as the trained model, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a time-frequency mask generation model configured to generate the time-frequency mask from a sound feature vector of the mixture signals and the text-embedded vector, and the sound extraction device is configured to calculate the sound feature vector of the mixture signals from the mixture signals, calculate, using the text-embedded extraction model, the text-embedded vector from the text representing the range of the extraction target sound, and generate, using the time-frequency mask generation model, the time-frequency mask from the calculated text-embedded vector and sound feature vector of the mixture signals.
4. The sound extraction system according to claim 2, wherein the storage device stores, as the trained model, a feature extraction model configured to output a sound feature vector of the mixture signals from the mixture signals, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a time-frequency mask generation model configured to generate the

time-frequency mask from the sound feature vector of the mixture signals and the text-embedded vector, and the sound extraction device is configured to calculate, using the feature extraction model, the sound feature vector of the mixture signals from the mixture signals, calculate, using the text-embedded extraction model, the text-embedded vector from the text representing the range of the extraction target sound, and generate, using the time-frequency mask generation model, the time-frequency mask from the calculated text-embedded vector and sound feature vector of the mixture signals.

5. The sound extraction system according to claim 1, wherein the learning device is configured to generate, as the trained model, a feature extraction model configured to output a sound feature vector of the mixture signals from the mixture signals, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a time-frequency mask generation model configured to generate the time-frequency mask from the sound feature vector of the mixture signals and the text-embedded vector, and the sound extraction device is configured to calculate, using the feature extraction model generated by the learning device, the sound feature vector of the mixture signals from the mixture signals, calculate, using the text-embedded extraction model generated by the learning device, the text-embedded vector from the text representing the range of the extraction target sound, and generate, using the time-frequency mask generation model generated by the learning device, the time-frequency mask from the calculated text-embedded vector and sound feature vector of the mixture signals.

6. The sound extraction system according to claim 1, further comprising: a learning device configured to generate a trained model, which is used for generating the time-frequency mask, by acquiring an initial trained model from an outside and updating the initial trained model by executing machine learning using a training data set including a target signal corresponding to the extraction target sound, training mixture signals obtained by mixing the target signal and a signal corresponding to noise other than the extraction target sound, and a training text corresponding to the extraction target sound, wherein the sound extraction device is configured to generate, using the trained model generated by the learning device, the time-frequency mask based on the mixture signals and the text representing the range of the extraction target sound.

7. The sound extraction system according to claim 1, further comprising: a storage device which stores a trained model used for extracting the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and the text representing the range of the extraction target sound, wherein the sound extraction device is configured to extract, using the trained model, the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and the text representing the range of the extraction target sound.

8. The sound extraction system according to claim 7, wherein the storage device stores, as the trained model, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a signal extraction model configured to generate a time waveform of the signal corresponding to the extraction target sound from the text-embedded vector and a time waveform of the mixture signals, and the sound extraction device is configured to calculate, using the text-embedded extraction model, the text-embedded vector from the text representing the range of the extraction target sound, and extract, using the signal extraction model, the signal corresponding to the extraction target sound from the mixture signals by generating the time waveform of the signal corresponding to the extraction target sound from the calculated text-embedded vector and the time waveform of the mixture signals.

9. The sound extraction system according to claim 7, further comprising: a learning device which generates a trained model, which is used for extracting the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and the text representing the range of the extraction target sound, by executing machine learning using a training data set

including a target signal corresponding to the extraction target sound, training mixture signals obtained by mixing the target signal and a signal corresponding to noise other than the extraction target sound, and a training text corresponding to the extraction target sound, wherein the sound extraction device is configured to extract, using the trained model generated by the learning device, the signal corresponding to the extraction target sound from the mixture signals, based on the mixture signals and the text representing the range of the extraction target sound.

10. The sound extraction system according to claim 7, further comprising: a learning device which generates a trained model, which is used for extracting the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and the text representing the range of the extraction target sound, by acquiring an initial trained model from an outside and updating the initial trained model by executing machine learning using a training data set including a target signal corresponding to the extraction target sound, training mixture signals obtained by mixing the target signal and a signal corresponding to noise other than the extraction target sound, and a training text corresponding to the extraction target sound, wherein the sound extraction device is configured to extract, using the trained model generated by the learning device, the signal corresponding to the extraction target sound from the mixture signals, based on the mixture signals and the text representing the range of the extraction target sound.

11. The sound extraction system according to claim 1, further comprising: a storage device in which the target signal corresponding to the extraction target sound and the signal corresponding to noise other than the extraction target sound are stored, wherein the learning device is configured to generate the training mixture signals by reading out the target signal corresponding to the extraction target sound and the signal corresponding to the noise other than the extraction target sound from the storage device and by mixing the target signal corresponding to the extraction target sound and the signal corresponding to the noise other than the extraction target sound.

12. The sound extraction system according to claim 1, wherein the text is a text representing onomatopoeia.

13. The sound extraction system according to claim 1, wherein the text is a text representing descriptive text.

14. A sound extraction method comprising: providing a sound extraction device configured to extract, from mixture signals including a signal corresponding to an extraction target sound, the signal corresponding to the extraction target sound; extracting the signal corresponding to the extraction target sound from the mixture signals based on the mixture signals and a text representing a range of the extraction target sound; generating, based on the mixture signals and the text representing the range of the extraction target sound, a time-frequency mask for extracting the signal corresponding to the extraction target sound; applying the time-frequency mask to the mixture signals and extract the signal corresponding to the extraction target sound from the mixture signals; generating, using a learning device, a trained model, which is used for generating the time-frequency mask, by executing machine learning using a training data set including a target signal corresponding to the extraction target sound, training mixture signals obtained by mixing the target signal and a signal corresponding to noise other than the extraction target sound, and a training text corresponding to the extraction target sound; generating, using the trained model generated by the learning device, the time-frequency mask based on the mixture signals and the text representing the range of the extraction target sound, wherein the learning device is configured to generate, as the trained model, a text-embedded extraction model configured to output a text-embedded vector from data obtained by preprocessing the text representing the range of the extraction target sound, and a time-frequency mask generation model configured to generate the time-frequency mask from a sound feature vector of the mixture signals and the text-embedded vector; calculating the sound feature vector of the mixture signals from the mixture signals; calculating, using the text-embedded extraction model generated by the learning device, the text-embedded vector from the text representing the range of the extraction target sound; and generating, using the time-frequency

mask generation model generated by the learning device, the time-frequency mask from the calculated text-embedded vector and sound feature vector of the mixture signals.
